CHUSHU TAN

Computational Neuroscience · Psychiatric Multi-Omics · AI for Science
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RESEARCH INTERESTS

- **Quantitative Electrophysiology & EEG Parameterization**: Spectral parameterization (FOOOF / specparam), aperiodic 1/f dynamics, Individual Alpha Frequency (IAF) stability, functional connectivity (oPLI), topological brain network construction (Orthogonal Minimum Spanning Tree [OMST]), leakage-free machine learning decoding.
- **Neuroimaging & Multimodal Evidence Synthesis**: Task-based fMRI, coordinate-based Activation Likelihood Estimation (ALE) meta-analysis, statistical power evaluation in neuroimaging, amyloid-β PET kinetic modeling (DVR vs. SUVR), CSF and plasma biomarker meta-analysis (PRISMA 2020 / GRADE).
- **Psychiatric Multi-Omics & Systems Genomics**: Major depressive disorder (MDD), mitochondrial oxidative phosphorylation (OXPHOS), cross-tissue blood and postmortem brain transcriptomics, paired Illumina EPIC 850K DNA methylation, GWAS summary-level synthesis (FUMA/MAGMA, SMR), Mendelian randomization.
- **Viral Oncogenomics & Tumor Microenvironment**: Epstein–Barr virus (EBV) genomic load stratification, nasopharyngeal carcinoma (NPC), bulk and single-cell RNA-seq (NPC43), myeloid-inflammatory polarization (*CXCL8*, *S100A8/A9*, *TREM1*), immune deconvolution.
- **AI for Scientific Discovery**: Autonomous LLM multi-agent architectures (LangChain, LangGraph), tool-augmented reasoning, dynamic API routing, automated and reproducible biomedical analytical workflows.

EDUCATION

- **Master of Applied Psychology (part-time)**
  China University of Political Science and Law (CUPL), Beijing, China | **2021 – present**
  Focus: Computational Psychiatry, Cognitive Neuroscience, Neuroimaging Evidence Synthesis
- **Master of Information Technology (M.IT)**
  Monash University, Melbourne, Australia | **2018 – 2019**
  Focus: Software Architecture, Distributed Database Systems, Machine Learning Algorithms
- **Bachelor of Science in Biological Sciences & Psychology (Double Major)**
  State University of New York at Buffalo (SUNY Buffalo), Buffalo, NY, USA | **2011 – 2017**
  Coursework: Molecular Genetics, Evolutionary Biology, Cognitive Psychology, Biopsychology, Biostatistics

PUBLICATIONS & WORKING MANUSCRIPTS

Peer-Reviewed Journal Articles

1. **Tan C.*** (2026). Depression and amyloid-β across CSF, PET, and plasma biomarkers: a systematic review and meta-analysis. ***Neuroscience***, 614, 82–93. DOI: 10.1016/j.neuroscience.2026.07.065. `[Solo First & Corresponding Author]`
2. Alsaadawe M.†, Radman B. A.†, Hu L., Long J., Luo Q., **Tan C.**, Amirat H. S., Alsaadawi M.*, & Lyu X.* (2025). From Viral Infection to Malignancy: The Dual Threat of EBV and COVID-19 in Cancer Development. ***Viruses***, 17(9), 1195. DOI: 10.3390/v17091195. `[Co-author, PMC12474468]`

Manuscripts Under Review & Submission Ready

3. **Tan C.*** (2026). Childhood Adversity and Reward-Related Brain Function in Task-Based Functional MRI: A Systematic Review and Activation Likelihood Estimation Meta-Analysis. *Under Review at* ***Child Abuse & Neglect*** (CHIABUNEG-S-26-02012). `[First & Corresponding Author]`

4. **Tan C.*** (2026). Parameterized Resting-State EEG Signatures of Acute Sleep Deprivation: Aperiodic Offset Elevation, Alpha Frequency Stability, and the Limits of Calibration-Free Decoding. *Target: **SLEEP*** (Completed Draft v15, submission package finalized). `[First & Corresponding Author]`
5. **Tan C.†**, Alsaadawe M.†, Hu T.†, & Lyu X.* (2026). Plasma EBV DNA Grouping Uncovers a Stage-Robust Myeloid-Inflammatory Signature in Nasopharyngeal Carcinoma. *Advanced Working Manuscript (Draft 12)*. `[Co-first Author]`
6. **Tan C.*** (2026). Reassessing Oxidative Phosphorylation Associations in Depression Using Public Transcriptomic, Methylomic and Genetic Data. *Full Pipeline Release (31 Reproducible Analysis Modules across 13 Cohorts)*. `[Lead Investigator]`

RESEARCH EXPERIENCE

Independent Researcher / Computational Biology Consultant

Remote | 2023.10 – present - **Resting-State EEG Parameterization & Sleep Deprivation Benchmark**: - Executed a reproducible neurophysiological investigation across OpenNeuro `ds004902` (N = 71 paired participants, 142 sessions; 100% QC retention). - Designed automated preprocessing using FastICA and spherical spline bad-channel repair. - Parameterized power spectra via `specparam/FOOOF`, demonstrating significant aperiodic broadband offset elevation (d_z = +0.788, $p < 10^{-8}$) and robust Individual Alpha Frequency (IAF) stability (ICC = 0.816). - Implemented strictly leakage-free Leave-One-Subject-Out (LOSO) machine learning classification (SVM AUC = 0.881, accuracy 83.1%) and continuous vigilance regression (r = 0.750, R^2 = 0.551). - Engineered a 19-ch surface Laplacian (CSD) harmonizer, Orthogonal Minimum Spanning Tree (OMST) topological graph builder, and within-site normative Z-score calibration framework. - **Multimodal Biomarker & Coordinate-Based fMRI Meta-Analyses**: - *Amyloid-β in Depression*: Conducted a 24-study meta-analysis synthesizing CSF, PET, and plasma biomarkers under PRISMA 2020 and GRADE frameworks, uncovering critical quantification divergence between DVR kinetic modeling and SUVR (published in *Neuroscience*, 2026). - *Childhood Adversity fMRI*: Built coordinate-based Activation Likelihood Estimation (ALE) meta-analyses across reward processing paradigms, quantifying literature-wide statistical power deficits and spatial convergence boundaries. - **MDD–OXPHOS Multi-Omics Re-analysis Framework**: - Engineered 31 automated R and Python pipelines processing 13 independent clinical cohorts ($N > 2,000$). - Synthesized peripheral blood transcriptomics (6 cohorts), postmortem brain transcriptomics (DLPFC, AnCg, nAcc, HIP), paired Illumina EPIC 850K methylation, and statistical genetics (GWAS, FUMA/MAGMA, SMR). - Rigorously re-evaluated previously asserted mitochondrial/OXPHOS dysfunction, revealing that past findings conflate medical comorbidity, medication exposure, and cellular composition shifts. - **Viral Oncology & Nasopharyngeal Carcinoma Microenvironment**: - Stratified recurrent/metastatic NPC cohorts by pre-treatment plasma EBV DNA copy number. - Integrated bulk transcriptomics, stage-adjusted sensitivity analysis, immune deconvolution (CIBERSORT, MCP-counter), and single-cell RNA-seq (NPC43), demonstrating a tumor-intrinsic myeloid-inflammatory signature (*CXCL8*, *S100A8/A9*, *TREM1*) in low-load tumors and adaptive exhaustion in high-load tumors. - **Interpretable Clinical Machine Learning**: - Formulated survival risk and classification models using XGBoost, Cox Proportional Hazards, and SHAP tree value attributions, resolving nonlinear feature interactions for clinical biomarkers.

AI / Bioinformatics Algorithm Engineer

Hisi Medical Technology, Guangzhou, China | 2026.01 – 2026.04 - **Scientific LLM Agent Collective**: Architected autonomous research agents using LangChain and LangGraph for high-throughput literature retrieval, dynamic tool-calling (NCBI, Ensembl, UniProt APIs), and multi-step reasoning. - **High-Throughput Genomic Pipelines**: Constructed automated analytical workflows integrating GWAS association analysis, TWAS expression prediction, and epigenetic functional element fine-mapping.

DATA ENGINEERING & INDUSTRY EXPERIENCE

Data Engineer

HSBC (via CEC outsourcing), Guangzhou, China | 2026.06 – present
Tech Stack: Prophecy, PySpark, Google Cloud Platform (GCP), Apache Airflow, Jenkins, SQL - Develop and optimize production financial ETL pipelines with Prophecy and PySpark on GCP. - Manage complex workflow dependencies via Apache Airflow; configure automated CI/CD pipelines via Jenkins.

Senior Data Analyst (contract)

HSBC, Guangzhou, China | 2024.03 – 2025.10
Tech Stack: OCR, NLP, Apache Spark, Hive, Airflow, LLM APIs, Python - Built automated financial disclosure parsing systems using OCR + NLP; integrated enterprise LLM APIs for unstructured text extraction, risk entity recognition, and sentiment classification. - Maintained large-scale ETL pipelines on Spark and Hive, performing data validation and performance tuning across high-throughput data tables.

Bioinformatics Engineer

Cloudna Biotech, Changsha, China | 2021.11 – 2023.10
Tech Stack: Nextflow, Snakemake, CWL, Docker, Linux, Python, R - Engineered automated WGS and WES pipelines with CWL, Nextflow, and Snakemake; containerized bioinformatics toolchains with Docker for deployment across distributed HPC clusters. - Built the DbABC resistome database web service (BLAST + RGI) and integrated multi-source databases into drug–disease knowledge graphs.

Bioinformatics Engineer / Software Developer

Xiansai / Sanji Biotech, Changsha, China | 2020.06 – 2021.10
Tech Stack: NGS Pipelines, Linux Shell, Python, MySQL - Built clinical sequencing analysis workflows for hemoglobinopathy variant detection; developed MySQL database management tools.

TECHNICAL & METHODOLOGICAL COMPETENCIES

- **Electrophysiology & Neuroimaging**: MNE-Python, FOOOF / specparam (spectral parameterization), Surface Laplacian (CSD), oPLI, OMST topological graph construction, LOSO cross-validation, GingerALE coordinate meta-analysis, PRISMA 2020, GRADE framework.
- **Genomics & Multi-Omics**: Bulk RNA-seq (DESeq2, limma), scRNA-seq / snRNA-seq (Scanpy, Seurat, Harmony, scVI), DNA Methylation (minfi, ChAMP), GWAS & TWAS (PLINK2, GEMMA, FUMA, SMR), Mendelian Randomization (TwoSampleMR), Immune Deconvolution (CIBERSORTx, MCP-counter).
- **Machine Learning & Statistics**: XGBoost, LightGBM, Random Forest, SVM, Lasso/Ridge/Elastic Net, CoxPH survival modeling, SHAP explainability, PyTorch, Linear Mixed-Effects Models (LMM), multiple testing correction (BH-FDR).
- **Big Data, Cloud & DevOps**: Python (NumPy, SciPy, Pandas, Scikit-learn), R / Bioconductor, SQL, Shell, Apache Spark / PySpark, Hive, Airflow, Hadoop, Prophecy, GCP, Docker, Nextflow, Snakemake, Jenkins CI/CD, Git / GitHub.
- **AI & Autonomous Systems**: LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Vector Databases (FAISS, Chroma), Tool-Calling, Prompt Engineering, OCR + NLP pipelines.

LANGUAGES & PROFESSIONAL ATTRIBUTES

- **Languages**: Mandarin Chinese (Native) | English (Full Professional Working Proficiency; US & Australian higher degrees, English-language peer-reviewed publications).
- **Open Science & Code**: Dedicated to reproducible research pipelines, open data sharing, and transparent statistical methodology.